CHARLIE LI

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EDUCATION

Bachelor of Science, Computer Science / Biology Second Major with Specialization in Genomics and Computational Biology / Chemistry Minor Expected: May 2027

Washington University in St. Louis

- **GPA:** 4.00/4.00
- **Honors:** Junior Academic Excellence Award (1 of 34 recipients in junior engineering class), Dean's List
- **Sample Coursework:** Geometric Computing for Biomedicine, AI for Chemistry, Biochemistry, Endocrinology, GPU Programming, Algorithms & Data Structures, Software Development

CLINICAL RESEARCH & EXPERIENCE

Hospice Volunteer Feb 2026 - Present

SSM Health

- Provided weekly bedside companionship to terminally ill patients, engaging in conversations to alleviate isolation.
- Identified and communicated patient needs condition changes to nursing staff to assist with comprehensive care plans.
- Assisted with patient requests and errands to ensure individual needs are met promptly.

Volunteer Crisis Counselor Jan 2026 - Present

Crisis Text Line

- Managed high-acuity crisis conversations via text, utilizing active listening and collaborative problem-solving to de-escalate situations involving suicide, self-harm, and domestic violence.
- Conducted real-time risk assessments to determine imminent danger.
- Co-created safety plans with texters, identifying coping mechanisms and social supports to bridge the gap between immediate crisis and long-term care.

Research Trainee May 2026 - Jul 2026

Advanced Summer Program for Investigation & Research Education (ASPIRE)

- Drive retrospective outcomes research for Merkel cell carcinoma by managing a 200+ patient REDCap database to analyze disease-specific survival and recurrence patterns.
- Evaluate the clinical efficacy of adjuvant radiation and immunotherapy while actively participating in multidisciplinary cutaneous oncology tumor boards to develop a primary manuscript.

Pediatric Emergency Medicine Research Associate Jan 2025 - Aug 2025

St. Louis Children's Hospital (PEMRAP) | St. Louis, MO

- Screened emergency department track boards to identify eligible candidates for active clinical trials.
- Recruited and consented over 300 patients and families for studies regarding resource availability.
- Collaborated with PIs and physicians to ensure research protocol adhered to clinical workflows and ethics compliance.

Emergency Department Shadowing & Cardiovascular Volunteer May 2024 - Aug 2025

Barnes Jewish Hospital | St. Louis, MO

- **Shadowing (Emergency Dept.):** Observed full cycle of emergency care, including triage, trauma stabilization, and discharge planning. Gained exposure to decision-making under high-pressure constraints and EHR utilization.
- **Volunteer (Cardiovascular Procedure Center):** Managed patient intake and discharge logistics, guided families to recovery bays, and acted as a liaison between nursing staff and families to stream communication during procedures.

RESEARCH EXPERIENCE

Research Assistant Aug 2025 - Present

Losos Lab (Evolutionary Biology) | St. Louis, MO

- Demonstrated deterministic host-microbiome co-evolution and proposed symbiont niche partitioning in *Anolis* lizards.
- Performed wet-lab extraction and 16S rRNA amplicon sequencing, processing reads and imputing functional abundances using QIIME 2 and PICRUSt2 in cloud environments.
- Built Random Forest classifiers and PERMANOVA models in R to classify microbiome variations, and applied hypervolume overlaps to quantify compositional similarities.

Research Assistant Jan 2024 - May 2025
Held Lab (Structural Biology) | St. Louis, MO

- Engineered a Python pipeline to analyze residue-scale solvent exposure dynamics across 18M+ PDB entries.
- Correlated protein crypticity with residue properties to profile structural behaviors at scale.
- Elucidated the redox opening mechanism of the UNC-49 GABA receptor via MD simulations, identifying Met 296 as the gating residue.
- Designed a custom trajectory-centering algorithm to overcome native MDAnalysis limitations, constructed free energy surfaces and discovered critical Cys238-Glu184 hydrogen bonds.

LEADERSHIP & SERVICE

Volunteer Birdkeeper May 2024 - Present
World Bird Sanctuary | St. Louis, MO

- Collaborated with rehabilitation staff during medical examinations of injured birds of prey, utilizing closed-loop communication to ensure team safety and secure handling.
- Managed daily husbandry, individualized nutritional preparation, and enclosure maintenance for rehabilitating raptors.
- Navigated the ethical and emotional complexities of treatment limitations and humane euthanasia.

Co-Chair (Question Writing Committee); Member (Community Outreach Committee) Aug 2023 - Present
WashU Stem Education Association | St. Louis, MO

- Directed a 10-student committee and coordinated with university professors to craft original, peer-reviewed chemistry questions for an annual national competition.
- Collaborate with ASEEDER to host the tournament in Shanghai, China.
- Developed an exam analytics website and an AI workflow to predict question difficulty, and restructured scoring software to support tiered divisions, increasing accessibility for under-resourced schools.
- Designed and presented interactive chemistry demos for students in underserved St. Louis area middle schools.
- Developed and delivered 1-hour science modules, translating complex chemical principles into accessible lessons to foster early engagement in STEM.
- Collaborated with educators to align hands-on experiments with existing science curricula, adapting teaching strategies to maximize student comprehension and retention.

FEATURED PROJECTS

Automated Neuron Segmentation from 2pFLIM Imaging
https://github.com/P-mandevillei/CSE-5504_Project

- Designed a web interface for batch segmentation of neuron somas from 2pFLIM images.
- Implemented classic geometric algorithms, including morphological operations, marching squares contouring, PCA and SVD-ICP alignment, and Laplacian deformation.
- Achieved 85 - 90% mean accuracy compared to manual segmentations.

Accelerating Molecular Dynamics on GPU
https://github.com/P-mandevillei/CUDA_Molecular_Dynamics

- Accelerated force computations with CUDA GPU kernels, achieving up to a 100x speedup for 10,000-particle systems.
- Maximized occupancy to 89% and optimized memory throughput via shared memory tiling, privatization, and juxtaposed traversal.
- Implemented the cutoff algorithm with a Compressed Sparse Row (CSR) structure, yielding up to a 1000x speedup.

Comparative Analysis of Trust in Traditional Chinese Medicine (TCM)
https://github.com/P-mandevillei/trust_in_tcm_opinion_mining

- Mined 18,038 social media posts to comparatively evaluate post-pandemic public trust in TCM, classifying patient attitudes by performing Aspect-Based Sentiment Analysis (ABSA) and fine-tuning multilingual language models.
- Designed robust statistical frameworks to measure community consensus and mitigate data noise.
- Uncovered nuanced attitudes toward TCM but significant cross-cultural distrust toward evidence-based diabetes treatments, informing potential strategies for targeted public health interventions.

SKILLS

- **Clinical & Lab:** BLS Provider (AHA), PCR, Gel Electrophoresis, DNA Extraction, Chromatography
- **Languages:** Mandarin (Native), Wu Chinese (Native), English (Advanced)
- **Computational:** Python, R, C/C++, CUDA, TypeScript, PyTorch, QIIME2, PICRUSt2, MDAnalysis, React, Git, Linux, AWS